

Observations & Interpretation of Figures 1–6

Companion to *Statistical Characterisation of a Large-Scale Software Bug Repository*. Every value below is computed from 196,633 unique Mozilla Bugzilla bug reports (BugsRepo, Zenodo 15004067, CC-BY 4.0), observation window 2021-01-27 to 2024-11-15.

How to use this sheet. Each figure has a **full** version (~70 words) and a **condensed** version (~35 words) for when the observation box is tight or your handwriting is large. Put the text into your own words before writing it on the poster — you will be asked about it at the poster session, and phrasing you chose yourself is phrasing you can defend. Every numerical value matches the tables printed on the A3 poster. The glossary at the end explains each statistical term in plain language.

FIGURE 1 · TABLE 1

Distribution of Comments per Bug Report

FULL

The distribution is severely right-skewed (skewness **7.04**, kurtosis **90.86**). The mean of **7.33** exceeds the median of **5**, confirming that the long tail pulls the average upward, so the median is the more reliable measure of a typical bug. Half of all bugs are resolved with between **3** and **8** comments (IQR = **5**), yet **8.74%** of reports exceed the Tukey upper fence of **15.5**, and one bug attracted **429** comments. Discussion effort is therefore highly concentrated: most bugs are settled quickly, while a small minority consumes a disproportionate share of developer attention.

CONDENSED

Strongly right-skewed (skewness **7.04**): mean **7.33** exceeds median **5**, so the median better represents a typical bug. Half need **3–8** comments, but **8.74%** are outliers and one reached **429** — effort is concentrated in a small minority.

n = **196,633** · mean **7.33** · median **5** · mode **4** · SD **10.49** · Q1 **3** · Q3 **8** · IQR **5** · P99 **52** · max **429** · skewness **7.04** · kurtosis **90.86**

FIGURE 2 · TABLE 2

Bug Resolution Time by Severity Level

FULL

Median resolution time rises monotonically as severity decreases — **1.21** days for S1, **13.54** for S2, **25.28** for S3 and **39.25** for S4 — so the most severe bugs are resolved roughly **32×** faster than the least severe. A Kruskal–Wallis test ($H = \mathbf{2,765.7}$, $df = 3$, $p < 0.001$) confirms the four medians are not all equal. The interquartile range also widens sharply, from **6.77** days for S1 to **207.54** for S4, showing that severe bugs are handled both faster and far more predictably, whereas low-severity bugs may be fixed within a week or linger for years.

CONDENSED

Median resolution time rises steadily from **1.21** days (S1) to **39.25** days (S4), a **32×** gap confirmed by Kruskal–Wallis ($p < 0.001$). The widening IQR (**6.77** → **207.54**) shows severe bugs are also handled far more predictably.

medians **1.21 / 13.54 / 25.28 / 39.25** days · IQR **6.77 / 52.35 / 138.04 / 207.54** · n = **428 / 4,876 / 19,056 / 33,153** · Kruskal–Wallis $H = \mathbf{2,765.7}$, $df = \mathbf{3}$, $p < \mathbf{0.001}$

FIGURE 3 · TABLE 3

Discussion Volume vs. Time to Resolution

FULL

The two variables are positively associated (Spearman $\rho = \mathbf{0.3175}$, $p < 0.001$), and the log–log slope of **0.909** indicates a near-proportional relationship. However, the coefficient of determination is only $r^2 = \mathbf{0.1254}$, meaning the model explains just **12.5%** of the variance in resolution time. With $n = \mathbf{6,000}$ the correlation is highly significant yet weak, which illustrates that statistical significance must not be mistaken for predictive strength. Comment count alone is therefore a poor predictor of how long a bug will take to close.

CONDENSED

Positive but weak association ($\rho = \mathbf{0.3175}$, $p < 0.001$); the log–log fit explains only **12.5%** of variance. Highly significant yet weak — significance is not predictive strength, and comment count alone cannot predict resolution time.

n = **6,000** · Spearman ρ **0.3175** · $p < \mathbf{0.001}$ · OLS slope **0.9085** · intercept **0.7172** · Pearson r **0.3542** · r^2 **0.1254**

FIGURE 4 · TABLE 4

Distribution of Bug Report Title Length

FULL

Title length is far more regular than the other variables examined: the mean of **9.69** words and median of **9** lie close together, and **50%** of all titles fall between **6** and **12** words. The distribution remains mildly right-skewed (**1.35**) with excess kurtosis **3.19**, and Shapiro–Wilk rejects normality ($W = \mathbf{0.9057}$, $p < 0.001$) — though with $n = \mathbf{4,000}$ even slight departures are detected, so $W \approx \mathbf{0.91}$ indicates a moderate rather than extreme deviation. This consistency suggests reporters follow an implicit convention when summarising a defect.

CONDENSED

Mean **9.69** and median **9** words are close, with **50%** of titles between **6** and **12**. Shapiro–Wilk rejects normality ($W = \mathbf{0.9057}$), but $W \approx 0.91$ signals only moderate deviation — reporters appear to follow an implicit convention.

n = **196,633** · mean **9.69** · median **9** · SD **4.87** · Q1 **6** · Q3 **12** · P90 **16** · P99 **26** · max **57** · skewness **1.35** · kurtosis **3.19** · Shapiro–Wilk W **0.9057** ($n = 4,000$)

FIGURE 5 · TABLE 5

Correlation Between Bug Report Attributes

FULL

All pairwise associations are weak to moderate; none exceeds $\rho = \mathbf{0.40}$. The strongest — CC-list size with comment count (**+0.400**) and resolution time with comment count (**+0.365**) — plausibly share an underlying cause: bugs that draw more people also draw more discussion and take longer to close. Title length is essentially uncorrelated with everything ($|\rho| \leq \mathbf{0.08}$), so report length carries no signal about a bug's complexity or lifetime. The absence of any strong correlation indicates that no single metadata field can substitute for the report text itself.

CONDENSED

No association exceeds $\rho = \mathbf{0.40}$. The strongest involve comment count (CC size **+0.400**, resolution time **+0.365**), likely sharing a common cause. Title length correlates with nothing ($|\rho| \leq \mathbf{0.08}$), so no metadata field substitutes for the report text.

CC size ~ comments **+0.400** · resolution time ~ comments **+0.365** · keywords ~ comments **+0.214** · keywords ~ resolution time **+0.212** · title length ~ all others $|\rho| \leq \mathbf{0.08}$ · method: Spearman rank correlation

FIGURE 6 · TABLE 6

Product and Component Concentration

FULL

Bug reports are distributed very unevenly across the project. A single product, **Core**, accounts for **41.2%** of all reports, and just **8** of the **85** products cover 80% of them. The component level is far more fragmented: the largest component, **General**, holds only **11.1%**, the top five reach just **22.4%** (against **71.0%** for the top five products), and **128** of the **725** components are needed to reach 80%. With a median component holding only **46** reports, this long tail is why automatic component assignment is a substantially harder classification problem than product assignment.

CONDENSED

Reports are highly concentrated by product — **Core** alone is **41.2%**, and **8** of **85** products cover 80%. Components are far more fragmented: **128** of **725** are needed for 80%, with a median of only **46** reports each.

products **85** · components **725** · Core **81,023 (41.2%)** · General **21,866 (11.1%)** · top-5 share **71.0%** (product) vs **22.4%** (component) · categories for 80%: **8** vs **128** · median size **127** vs **46**

Plain-language glossary — for answering questions

Mean vs. Median

Mean is the average; median is the middle value. *If four people earn ₹20k and one earns ₹10 lakh, the mean says ₹2.16 lakh but the median says ₹20k. The median describes a typical person better.*

Skewness

Whether the tail stretches right or left. Positive = a few very large values. *Most bugs get 5 comments; one got 429. That long right tail is skewness 7.04.*

Kurtosis

How extreme the outliers are. High kurtosis = rare but enormous values.

Quartiles & IQR

Sort the data and cut it into four equal parts. Q1 = 25% mark, Q3 = 75% mark; IQR = Q3 – Q1 = the range of the middle half. *Q1 = 3, Q3 = 8 means half of all bugs sit between 3 and 8 comments.*

Percentile

P90 = the value 90% of the data falls below. *P99 = 52 comments means 99% of bugs got fewer than 52.*

Box plot

The box is Q1→Q3, the line inside is the median, the whiskers reach the normal range, and the dots beyond are outliers. A tall box means the group is unpredictable.

Outlier (Tukey's rule)

Anything above Q3 + 1.5 × IQR counts as unusually large. *Here that is 15.5 comments — 8.74% of bugs exceed it.*

Correlation (Spearman ρ)

Do two things rise together? +1 = perfectly together, 0 = unrelated, –1 = opposite. We use Spearman (rank-based) instead of Pearson because our data is very lopsided. *Taller people are usually heavier — a positive correlation, but not a perfect one.*

r^2 (coefficient of determination)

How much of the variation one variable explains in the other. *$r^2 = 0.125$ means comment count explains 12.5% of why bugs take different times — the other 87.5% is something else.*

p-value & significance

The chance the pattern is a fluke. $p < 0.001$ = almost certainly real. *Careful: "real" is not "strong". With 6,000 data points even a faint pattern is detectable — which is exactly the point Figure 3 makes.*

Kruskal–Wallis test

Checks whether several groups have the same middle value. *Do S1, S2, S3 and S4 bugs really take different times, or is it chance? Answer: really different.*

Shapiro–Wilk test

Checks whether data follows the bell curve. W near 1 = bell-shaped. *Our $W = 0.91$ — close to a bell curve, but not exactly one.*

Log scale

Each step multiplies by 10 instead of adding. Used when values range from minutes to years, so small and huge values both stay visible on one axis.

KDE (kernel density estimate)

A smoothed version of the histogram — the same shape drawn as a curve instead of bars.